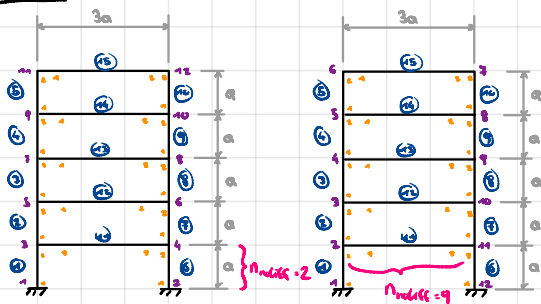


Problem 1)



$$EI = 1e^4 \quad \mu = 1e6 \quad E = 1e^4 \quad I_v = 1 \quad I_h = 2$$

using a high $\mu \rightarrow$ high $A \rightarrow$ high axial stiffness $\rightarrow$ negligible

$$\mu = \frac{AL^2}{I} \Rightarrow A = \frac{\mu I}{L^2}$$

horizontal beams:

$$A_h = \frac{\mu I_h}{9a^2}$$

vertical beams:

$$A_v = \frac{\mu I_v}{a^2}$$

a) x at node 11 = $r[10 \cdot 3] = 0,070836 \text{ m} = v_{11x}$

b) x at node 6 = $r[5 \cdot 3] = 0,070836 \text{ m} = v_{6x}$

c) $b_w = n_{\text{nodes}} = (n_{\text{diff}} + 1)$
 $n_{\text{diff}} = 3$

Scheme 1:

$$n_{\text{diff}} = 2$$

$$b_w = 3 \cdot (2 + 1) = 9$$

SCHEME 1

Scheme 2:

$$n_{\text{diff}} = 9$$

$$b_w = 3 \cdot (9 + 1) = 30$$

SCHEME 2

The first numbering scheme has a more band limited stiffness matrix (smaller b_w) and is therefore faster to solve.

(look at code)

d) $v_{11x} = r[10 \cdot 3] = 0,595239 \text{ m}$

new element at	v_{11x} / m	new element at	v_{11x} / m
1-4	0,05683	3-2	0,05683
3-6	0,04499	5-4	0,04499
5-8	0,04188	7-6	0,04188
7-10	0,04283	9-8	0,04283
9-12	0,05195	11-10	0,05185

adding an element either at

5-8 or 7-6 minimizes displacement at 11

f) hinge connection at 11:

no bending moment transfer between 3 and 11.

$$= \frac{3EI}{\ell^3} \begin{bmatrix} 1 & -\ell & -1 \\ -\ell & \ell^2 & \ell \\ -1 & \ell & 1 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{\theta 1} \\ v_{22} \end{bmatrix} = \frac{EI}{L^3} \begin{bmatrix} 3-3L & -3 \\ -3L & 3L^2 & 3L \\ -3 & 3L & 3 \end{bmatrix} \begin{bmatrix} v_{21} \\ v_{\theta 1} \\ v_{22} \end{bmatrix}$$

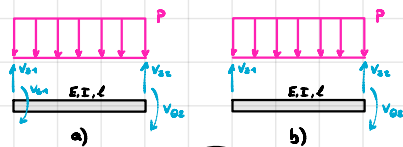
$$S = \begin{bmatrix} S_{x1} \\ S_{z1} \\ S_{\theta 1} \\ S_{x2} \\ S_{z2} \\ S_{\theta 2} \end{bmatrix} = \frac{EI}{L^3} \begin{bmatrix} \mu & 0 & 0 & -\mu & 0 & 0 \\ 0 & 12 & -6L & 0 & -12 & -6L \\ 0 & -6L & 4L^2 & 0 & 6L & 2L^2 \\ \mu & 0 & 0 & \mu & 0 & 0 \\ 0 & 6L & 2L^2 & 0 & 12 & 6L \\ 0 & -6L & 2L^2 & 0 & 6L & 4L^2 \end{bmatrix} \begin{bmatrix} v_{x1} \\ v_{z1} \\ v_{\theta 1} \\ v_{x2} \\ v_{z2} \\ v_{\theta 2} \end{bmatrix} = kv$$

where $\mu = AL^2/I$

x at node 11 = $r[10 \cdot 3] = 0,081195 \text{ m} = v_{11x}$

$$k_{\text{hinge}}^{(1)} = \frac{EI}{L^3} \begin{bmatrix} \mu & 0 & 0 & -\mu & 0 & 0 \\ 0 & 12 & -6L & 0 & -12 & -6L \\ 0 & -6L & 4L^2 & 0 & 6L & 2L^2 \\ \mu & 0 & 0 & \mu & 0 & 0 \\ 0 & 6L & 2L^2 & 0 & 12 & 6L \\ 0 & -6L & 2L^2 & 0 & 6L & 4L^2 \end{bmatrix}$$

Problem 2:



$$\underline{k}_{\text{elem}} = \frac{EI}{l^3} \begin{bmatrix} 12 & -6l & -12 & -6l \\ -6l & 4l^2 & 6l & 2l^2 \\ -12 & 6l & 12 & 6l \\ -6l & 2l^2 & 6l & 4l^2 \end{bmatrix} \begin{bmatrix} v_{z1} \\ v_{\theta 1} \\ v_{z2} \\ v_{\theta 2} \end{bmatrix} \quad \mathbf{v}_r = \begin{bmatrix} v_{z1} \\ v_{\theta 1} \\ v_{z2} \\ v_{\theta 2} \end{bmatrix} \quad \mathbf{v}_c = \begin{bmatrix} v_{z1} \end{bmatrix}$$

$$\underline{k}_{\text{elem}}^{\text{mod}} = \frac{EI}{l^3} \begin{bmatrix} 12 & -12 & -6l & -6l \\ -12 & 12 & 6l & 6l \\ -6l & 6l & 4l^2 & 2l^2 \\ -6l & 6l & 2l^2 & 4l^2 \end{bmatrix} \begin{bmatrix} v_{z1} \\ v_{\theta 1} \\ v_{z2} \\ v_{\theta 2} \end{bmatrix}$$

k_{rr} (top-left 2x2), k_{rc} (top-right 2x1), k_{cr} (bottom-left 1x2), k_{cc} (bottom-right 1x1)

$$\begin{aligned} \underline{k}_{rr}^m &= (k_{rr} - k_{rc} k_{cc}^{-1} k_{cr}) = \frac{EI}{l^3} \left(\begin{bmatrix} 12 & -12 & -6l \\ -12 & 12 & 6l \\ -6l & 6l & 4l^2 \end{bmatrix} - \begin{bmatrix} -6l \\ 6l \\ 2l^2 \end{bmatrix} \frac{1}{4l^2} \begin{bmatrix} -6l & 6l & 2l^2 \end{bmatrix} \right) \\ &= \frac{EI}{l^3} \left(\begin{bmatrix} 12 & -12 & -6l \\ -12 & 12 & 6l \\ -6l & 6l & 4l^2 \end{bmatrix} - \frac{1}{4l^2} \begin{bmatrix} 36l^2 & -36l^2 & -12l^3 \\ -36l^2 & 36l^2 & 12l^3 \\ -12l^3 & 12l^3 & 4l^4 \end{bmatrix} \right) \\ &= \frac{EI}{l^3} \left(\begin{bmatrix} 12 & -12 & -6l \\ -12 & 12 & 6l \\ -6l & 6l & 4l^2 \end{bmatrix} - \begin{bmatrix} 9 & -9 & -3l \\ -9 & 9 & 3l \\ -3l & 3l & l^2 \end{bmatrix} \right) = \frac{EI}{l^3} \begin{bmatrix} 3 & -3 & -3l \\ -3 & 3 & 3l \\ -3l & 3l & 3l^2 \end{bmatrix} = \underline{k}_{rr}^m \end{aligned}$$

$$\begin{aligned} \underline{\tilde{S}}^m &= \frac{Pl}{12} \begin{bmatrix} -6 \\ -6 \\ -l \end{bmatrix} \\ \underline{\tilde{S}}^{\text{mod}} &= \frac{Pl}{12} \begin{bmatrix} -6 \\ -6 \\ -l \end{bmatrix} \end{aligned}$$

$$k_{rc} k_{cc}^{-1} \underline{S}_c^m = \frac{EI}{l^3} \begin{bmatrix} -6l \\ 6l \\ 2l^2 \end{bmatrix} \frac{1}{4l^2} \frac{Pl}{12} = \begin{bmatrix} -6l \\ 6l \\ 2l^2 \end{bmatrix} \frac{P}{48} = \begin{bmatrix} -\frac{Pl}{8} \\ \frac{Pl}{8} \\ \frac{Pl^2}{24} \end{bmatrix}$$

$$\underline{\tilde{S}}^{\text{mod}} = \underline{\tilde{S}}_r^m - k_{rc} k_{cc}^{-1} \underline{S}_c^m = \frac{Pl}{12} \begin{bmatrix} -6 \\ -6 \\ -l \end{bmatrix} - \begin{bmatrix} -\frac{Pl}{8} \\ \frac{Pl}{8} \\ \frac{Pl^2}{24} \end{bmatrix} = \frac{Pl}{24} \begin{bmatrix} -9 \\ -5 \\ -l \end{bmatrix} = \frac{Pl}{8} \begin{bmatrix} -3 \\ -5 \\ -l \end{bmatrix}$$